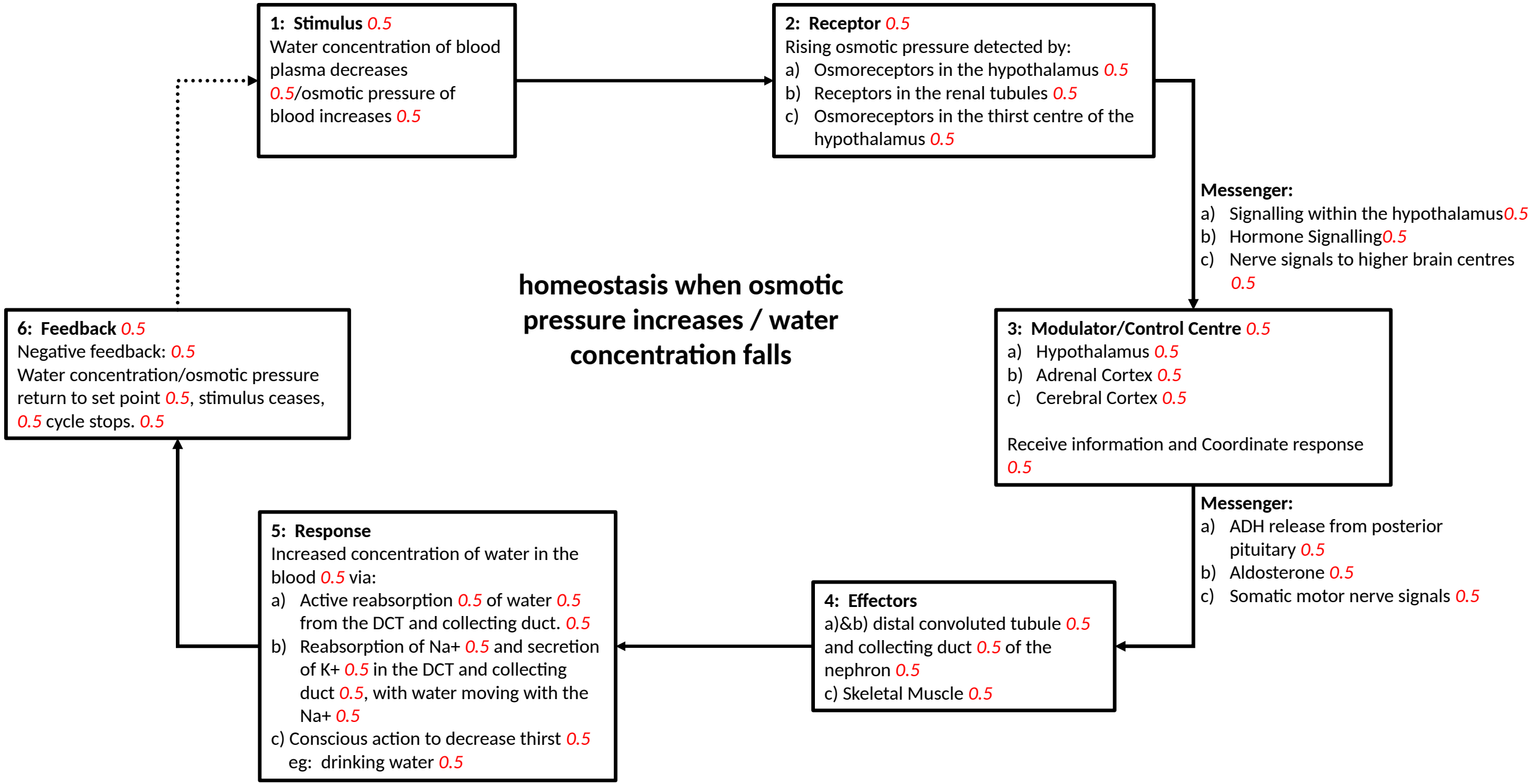

1: Stimulus 0.5
Water concentration of blood plasma decreases 0.5/osmotic pressure of blood increases 0.5
2: Receptor 0.5
Rising osmotic pressure detected by:
a) Osmoreceptors in the hypothalamus 0.5
b) Receptors in the renal tubules 0.5
c) Osmoreceptors in the thirst centre of the hypothalamus 0.5
Messenger:
a) Signalling within the hypothalamus0.5
b) Hormone Signalling0.5
c) Nerve signals to higher brain centres 0.5
3: Modulator/Control Centre 0.5
a) Hypothalamus 0.5
b) Adrenal Cortex 0.5
c) Cerebral Cortex 0.5
Receive information and Coordinate response 0.5
Messenger:
a) ADH release from posterior pituitary 0.5
b) Aldosterone 0.5
c) Somatic motor nerve signals 0.5
4: Effectors
a)&b) distal convoluted tubule 0.5 and collecting duct 0.5 of the nephron 0.5
c) Skeletal Muscle 0.5
5: Response
Increased concentration of water in the blood 0.5 via:
a) Active reabsorption 0.5 of water 0.5 from the DCT and collecting duct. 0.5
b) Reabsorption of Na+ 0.5 and secretion of K+ 0.5 in the DCT and collecting duct 0.5, with water moving with the Na+ 0.5
c) Conscious action to decrease thirst 0.5
eg: drinking water 0.5
6: Feedback 0.5
Negative feedback: 0.5
Water concentration/osmotic pressure return to set point 0.5, stimulus ceases, 0.5 cycle stops. 0.5
homeostasis when osmotic pressure increases / water concentration falls

Homeostasis when CO_2 levels rise above set point

1: Stimulus
Rising blood CO_2 levels/rising H+ levels (falling pH)
Due to:
- Increased exertion and cellular respiration
- Decreased rate and depth of breathing

2: Receptors
a) Rising CO_2 levels detected by Chemoreceptors in the aortic body and carotid arch
b) Rising H+ levels (falling pH) detected by chemoreceptors in the carotid arch and medulla oblongata

Messenger:
Nervous System

3: Modulator/Control Centre

Respiratory centre in the medulla oblongata receives information and coordinates actions to reduce CO_2 concentration

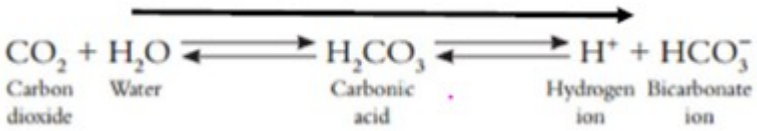
Messenger:
Autonomic Nervous System

4: Effectors
Diaphragm
Intercostal Muscles

5: Response
Diaphragm and Intercostal muscles act to increase rate and depth of breathing.

More CO_2 is excreted via the respiratory tract, so blood CO_2 falls, H+ concentration falls (pH rises)

6: Feedback
Negative feedback:
CO_2/H+/pH levels return to set point
Stimulus Ceases
Cycle Stops



Homeostasis when CO_2 levels fall below set point

1: Stimulus
Falling blood CO_2 levels/falling H+ levels (rising pH)
Due to:
- Hyperventilation (due to pain, anxiety, or voluntary choice)

2: Receptors
a) Falling CO_2 levels detected by Chemoreceptors in the aortic body and carotid arch
b) Falling H+ levels (rising pH) detected by chemoreceptors in the carotid arch and medulla oblongata

Messenger:
Nervous System

3: Modulator/Control Centre

Respiratory centre in the medulla oblongata receives information and coordinates actions to increase CO_2 concentration

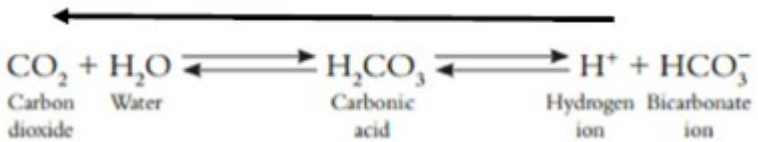
Messenger:
Autonomic Nervous System

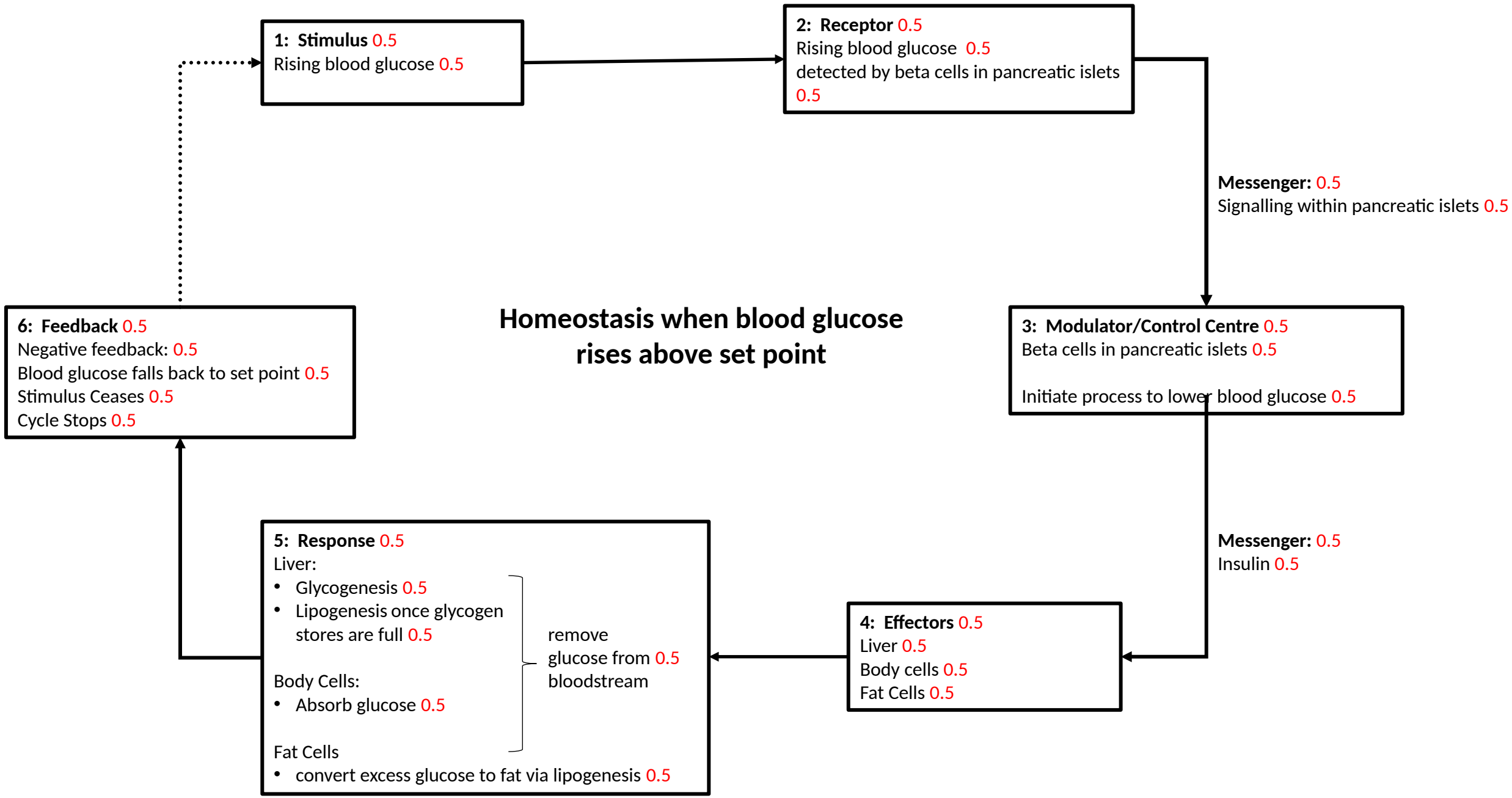
4: Effectors
Diaphragm
Intercostal Muscles

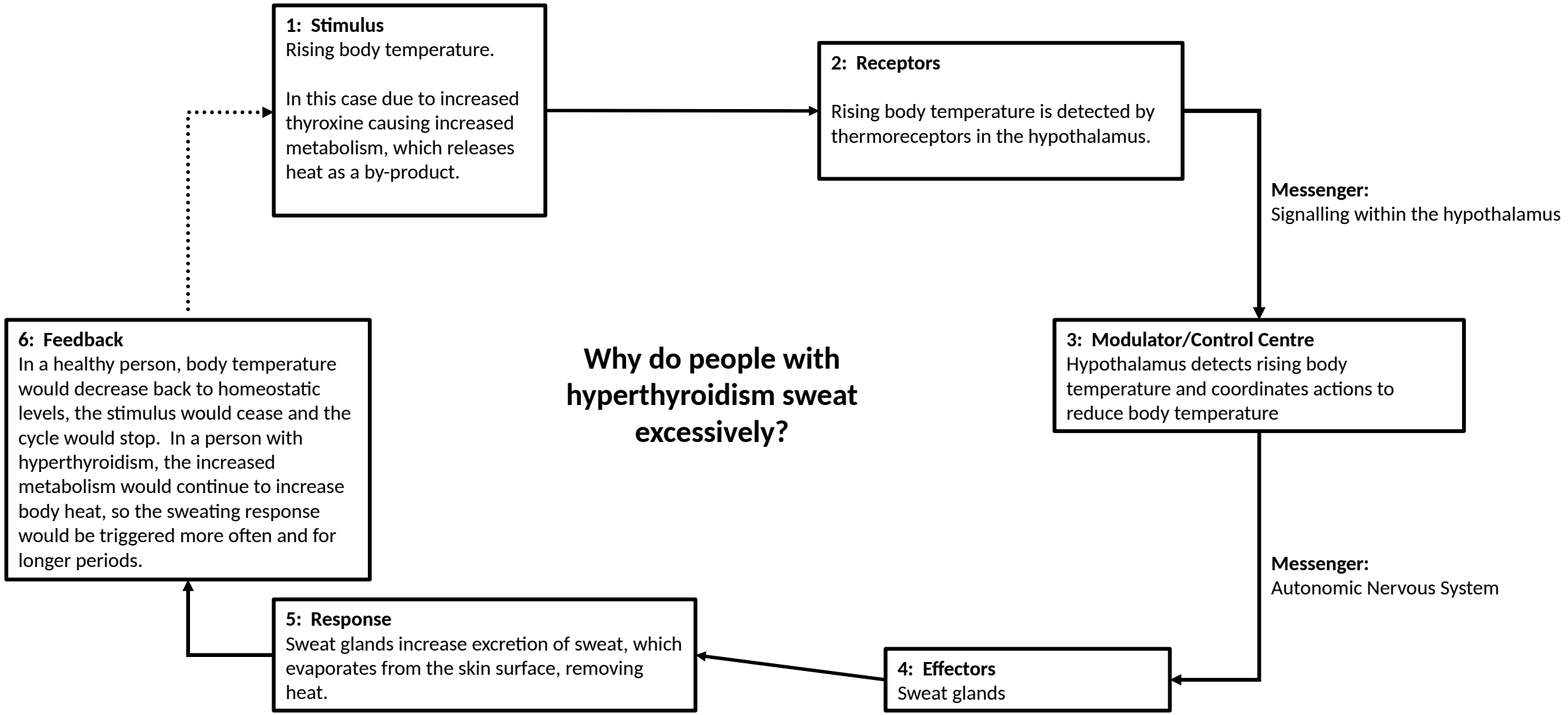
5: Response
Diaphragm and Intercostal muscles act to decrease rate and depth of breathing.

Less CO_2 is excreted via the respiratory tract, so blood CO_2 rises, H+ concentration rises (pH falls)

6: Feedback
Negative feedback:
CO_2/H+/pH levels return to set point
Stimulus Ceases
Cycle Stops




Homeostasis when blood glucose rises above set point
1: Stimulus 0.5
Rising blood glucose 0.5
2: Receptor 0.5
Rising blood glucose 0.5
detected by beta cells in pancreatic islets 0.5
Messenger: 0.5
Signalling within pancreatic islets 0.5
3: Modulator/Control Centre 0.5
Beta cells in pancreatic islets 0.5
Initiate process to lower blood glucose 0.5
Messenger: 0.5
Insulin 0.5
4: Effectors 0.5
Liver 0.5
Body cells 0.5
Fat Cells 0.5
5: Response 0.5
Liver:
• Glycogenesis 0.5
• Lipogenesis once glycogen stores are full 0.5
Body Cells:
• Absorb glucose 0.5
Fat Cells
• convert excess glucose to fat via lipogenesis 0.5
remove glucose from 0.5 bloodstream
6: Feedback 0.5
Negative feedback: 0.5
Blood glucose falls back to set point 0.5
Stimulus Ceases 0.5
Cycle Stops 0.5


1: Stimulus
Rising body temperature.
In this case due to increased thyroxine causing increased metabolism, which releases heat as a by-product.
2: Receptors
Rising body temperature is detected by thermoreceptors in the hypothalamus.
Messenger:
Signalling within the hypothalamus
3: Modulator/Control Centre
Hypothalamus detects rising body temperature and coordinates actions to reduce body temperature
Messenger:
Autonomic Nervous System
4: Effectors
Sweat glands
5: Response
Sweat glands increase excretion of sweat, which evaporates from the skin surface, removing heat.
6: Feedback
In a healthy person, body temperature would decrease back to homeostatic levels, the stimulus would cease and the cycle would stop. In a person with hyperthyroidism, the increased metabolism would continue to increase body heat, so the sweating response would be triggered more often and for longer periods.
Why do people with hyperthyroidism sweat excessively?


1: Stimulus (0.5)
Rising body temperature. (0.5)
In this case due to increased thyroxine (0.5) causing increased metabolism (0.5), which releases heat as a by-product. (0.5)
2: Receptor (0.5)
Rising body temperature is detected by thermoreceptors (0.5) in the hypothalamus. (0.5)
Messenger:
Signalling within the hypothalamus
3: Modulator/Control Centre (0.5)
Hypothalamus(0.5) detects rising body temperature (0.5) and coordinates actions to reduce body temperature (0.5)
Messenger:
Autonomic Nervous System
4: Effectors (0.5)
Sweat glands (0.5)
5: Response (0.5)
Sweat glands increase excretion of sweat (0.5), which evaporates from the skin surface (0.5), removing heat. (0.5)
6: Feedback (0.5)
In a healthy person, body temperature would decrease(0.5) back to homeostatic levels (0.5), the stimulus would cease (0.5)
and the cycle would stop. (0.5)
In a person with hyperthyroidism, the increased metabolism (0.5)
would continue to increase body heat(0.5), so the sweating response would be triggered more often and for longer periods. (0.5)
Why do people with hyperthyroidism sweat excessively?

Why is increased Respiratory Rate a symptom in people with emphysema?

1: Stimulus (0.5)
Rising blood CO_2 levels (0.5) /rising H+ levels (falling pH) (0.5)

In this case due to inability of the alveoli (0.5) to effectively remove CO2 from the blood (0.5)
to be excreted via the respiratory tract. (0.5)

2: Receptors (0.5)
a) Rising CO_2 levels detected by Chemoreceptors (0.5) in the aortic body and carotid arch (0.5)
b) Rising H+ levels (falling pH) detected by chemoreceptors in the carotid arch and medulla oblongata (0.5)

Messenger:
Nervous System

3: Modulator/Control Centre (0.5)

Respiratory centre (0.5) in the medulla oblongata (0.5) receives information and coordinates actions to reduce CO_2 concentration (0.5)

Messenger:
Autonomic Nervous System

4: Effectors (0.5)
Diaphragm (0.5)
Intercostal Muscles (0.5)

5: Response
Diaphragm and Intercostal muscles act to increase (0.5) rate and depth of breathing (0.5).

More CO_2 is excreted via the respiratory tract, so blood CO_2 falls, H+ concentration falls (pH rises) (0.5)

6: Feedback
In a person with healthy lungs, (0.5) CO_2/H+/pH levels return to set point (0.5), stimulus ceases (0.5)and the cycle stops (0.5) due to negative feedback. (0.5)

In a person with emphysema(0.5), CO2 levels will continue to be high (0.5) as the person will be unable to excrete it effectively (0.5)
even when respiratory rate increases. (0.5)

Homeostasis when CO_2 levels rise above set point

1: Stimulus
Rising blood CO_2 levels/rising H+ levels (falling pH)
Due to:
- Increased exertion and cellular respiration
- Decreased rate and depth of breathing

2: Receptors
a) Rising CO_2 levels detected by Chemoreceptors in the aortic body and carotid arch
b) Rising H+ levels (falling pH) detected by chemoreceptors in the carotid arch and medulla oblongata

Messenger:
Nervous System

3: Modulator/Control Centre

Respiratory centre in the medulla oblongata receives information and coordinates actions to reduce CO_2 concentration

Messenger:
Autonomic Nervous System

4: Effectors
Diaphragm
Intercostal Muscles

5: Response
Diaphragm and Intercostal muscles act to increase rate and depth of breathing.

More CO_2 is excreted via the respiratory tract, so blood CO_2 falls, H+ concentration falls (pH rises)

6: Feedback
Negative feedback:
CO_2/H+/pH levels return to set point
Stimulus Ceases
Cycle Stops

